SPHEREx Pipeline: Source Decontamination and Spectral Extraction

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Repository:
https://github.com/andrijazupic/SPHEREx_pipeline

Mission Overview: SPHEREx

- **Mission**:
  - Spectro-Photometer for the History of the Universe, Epoch of Reionization, and Ices Explorer
- **Type**:
  - NASA Near-Infrared All-Sky Spectral Survey
- **Orbit**:
  - Sun-synchronous Earth orbit (terminator orbit)
- **Survey Method**:
  - All-sky survey repeated 4 times over 2 years

Instrumental Specifications

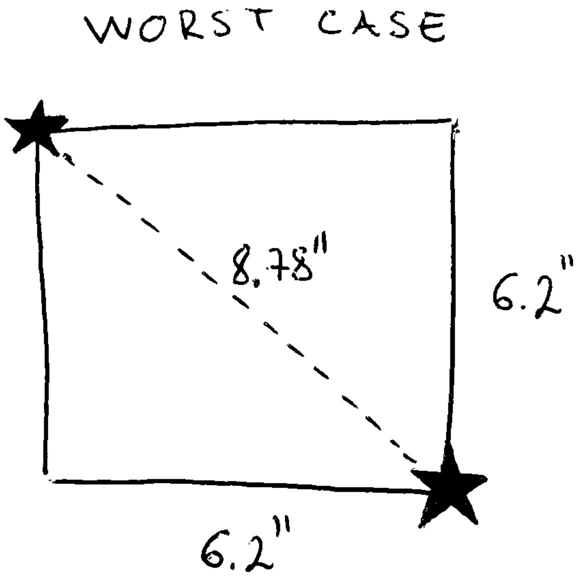
- **Wavelength Range**:
  - 0.75 to 5.0 µm
- **Spectral Resolution**:
  - $R \approx 41$ (for 0.75 - 2.42 µm)
  - $R \approx 135$ (for 2.42 - 5.0 µm)
- **Spatial Resolution**:
  - 6.2" per pixel
- **Observing Strategy (Spectro-Photometry):**
  - SPHEREx does not use a traditional spectrograph (like a grating or prism). Instead, it uses **Linear Variable Filters (LVFs)**
  - **The Mechanism**: As the satellite scans the sky, a target moves physically across the focal plane, sequentially passing through **~102 distinct narrow-band filter segments**.
  - **The Result**: By combining these ~102 separate photometric measurements of the same object taken at different times, we construct a **low-resolution spectrum**. This is why it is technically "spectrophotometry."

The Challenge: Large Pixel Scale

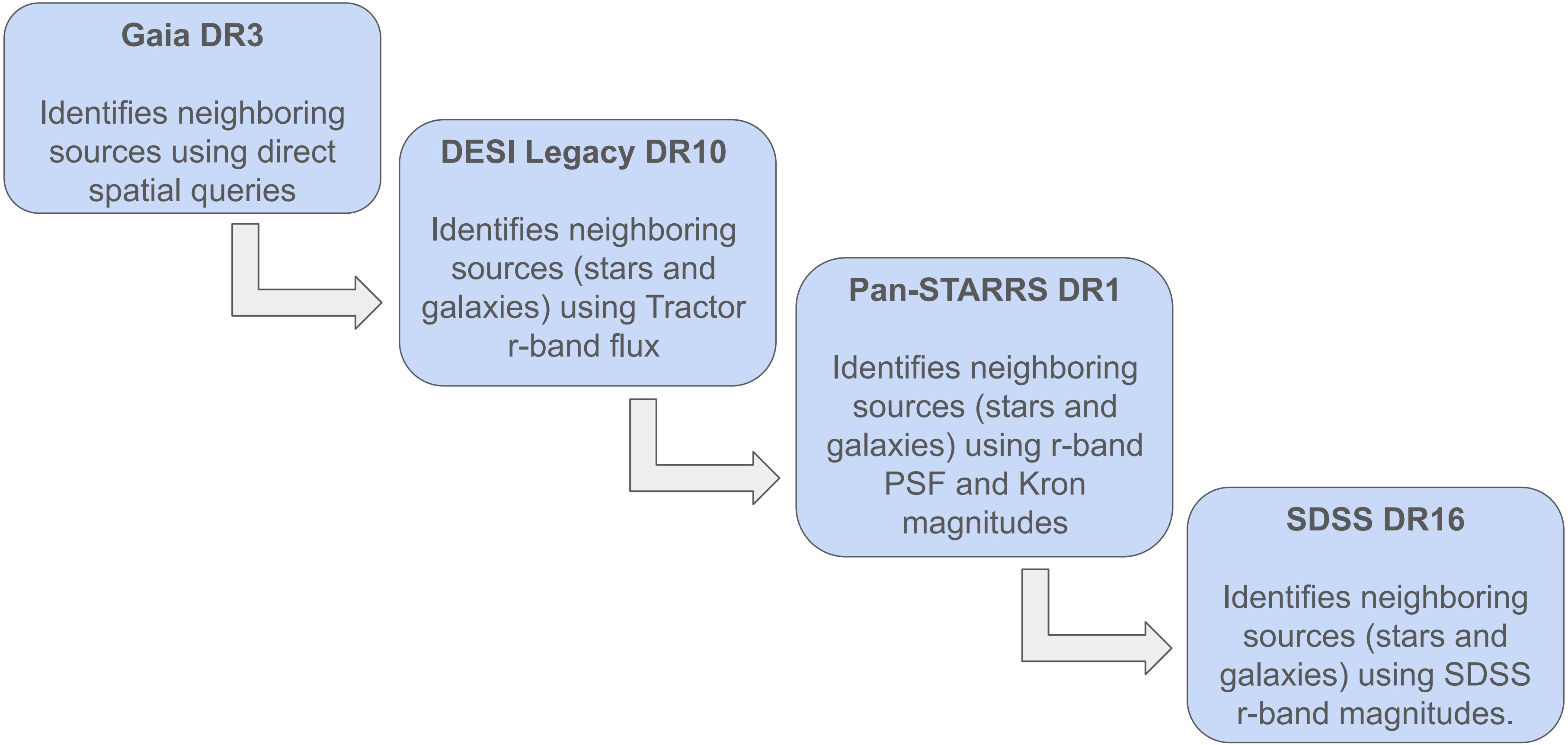
- **The Problem**:
  - SPHEREx pixels (6.2") are significantly larger than typical optical/NIR survey pixels
  - **Pixel Size / Resolution Comparison**:
    - Gaia: ~0.1" (effective angular pixel size)
    - Euclid: 0.1" (VIS visible) / 0.3" (NISP infrared)
    - Pan-STARRS: 0.25"
    - DESI Legacy: 0.26"
    - SDSS: 0.40"
    - Hipparcos: ~1.2" (focal plane grid period; note: used a photomultiplier, not modern CCD pixels)
    - **SPHEREx: 6.2"**
- **Consequence**:
  - A single SPHEREx pixel often contains multiple **unresolved sources**
- **Requirement**:
  - We must identify and **flag "contaminated" targets** where background flux can significantly bias the target spectrum

Multi-Survey Catalog Filtering

- **Catalog-Level Cleaning**:
  - We identify potential contaminants by cross-referencing our SPHEREx targets against pre-computed, high-resolution source catalogs within a radius of 9.3”
- **Why 9.3” radius?**
  - Because the exact sub-pixel position of a target is unknown, this radius accounts for the **worst-case geometric scenario** where a target and a contaminant fall in opposite corners of the same 6.2" SPHEREx pixel (an 8.78" diagonal separation), plus an **additional ~0.5" buffer** to capture the overlapping light from their Point Spread Functions (PSF).



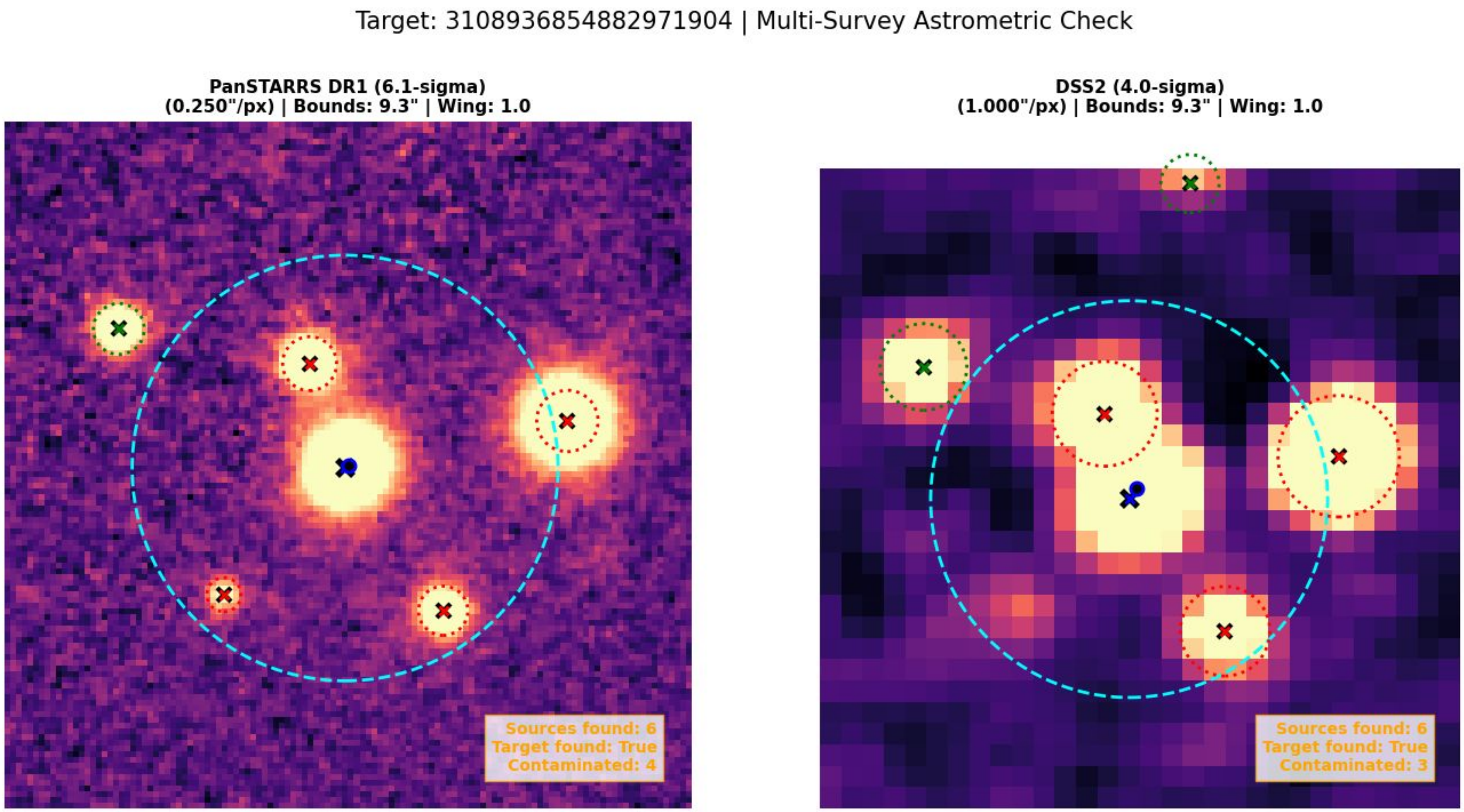
Multi-Survey Catalog Filtering



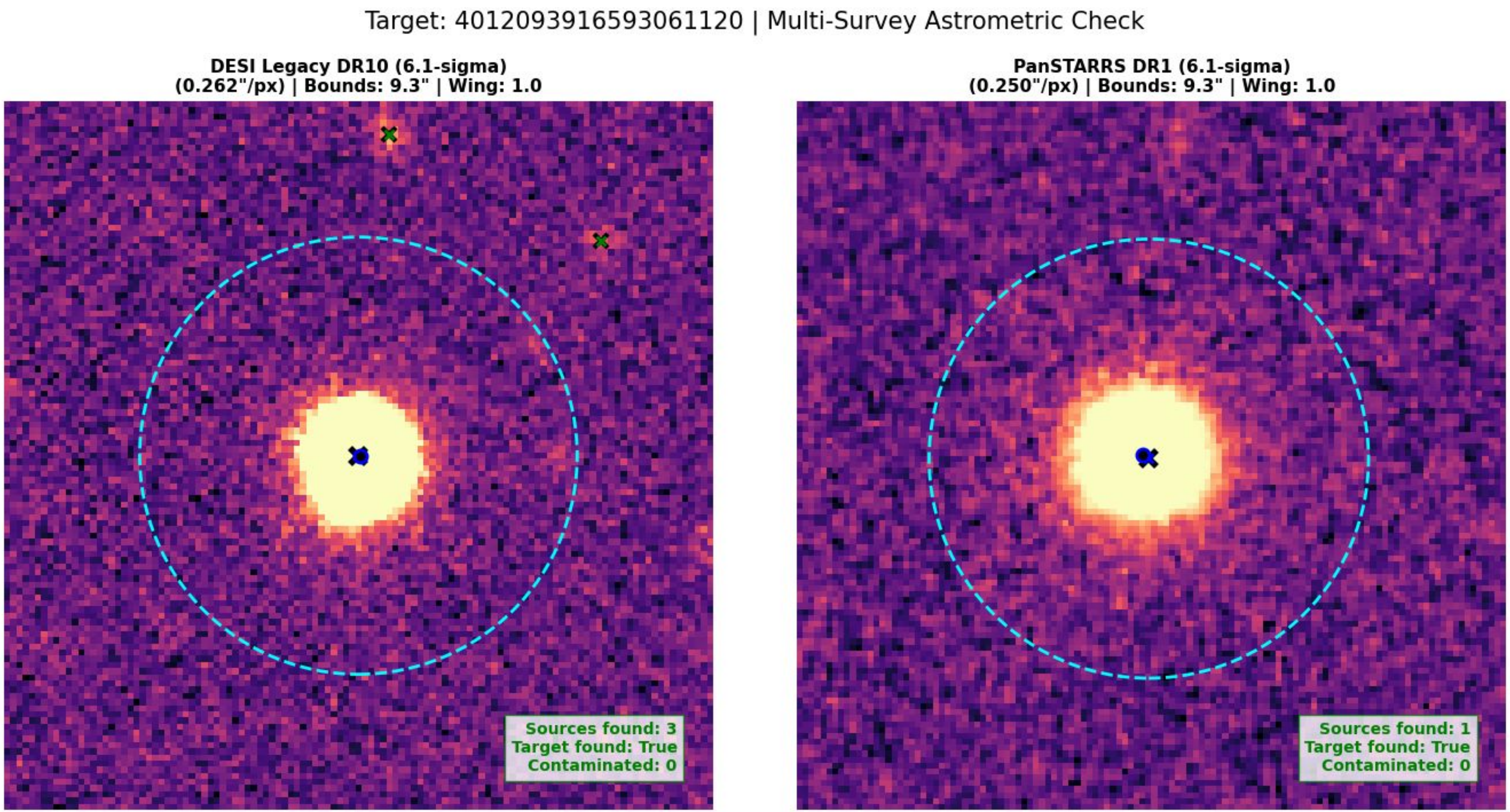
Multi-Survey Image Filtering

- For high-confidence cleaning, the code performs direct **image-level checks**:
  - **Fallback Hierarchy**: Sequentially fetches the highest-resolution optical FITS cutout available (Legacy ➡ Pan-STARRS ➡ SDSS ➡ DSS2)
  - **PSF Fitting**: Detects all sources within the 30" Field of View (FOV) using DAOStarFinder
  - **Exclusion Zone**: Evaluates all neighboring detections within the 9.3" radius around the target center.
  - **Dynamic PSF Wings**: Models the physical extent of neighbor light to determine if it bleeds into the target’s 9.3" extraction zone.

Multi-Survey Image Filtering



Multi-Survey Image Filtering



Spectral Extraction: SPXQuery vs. SPIFF

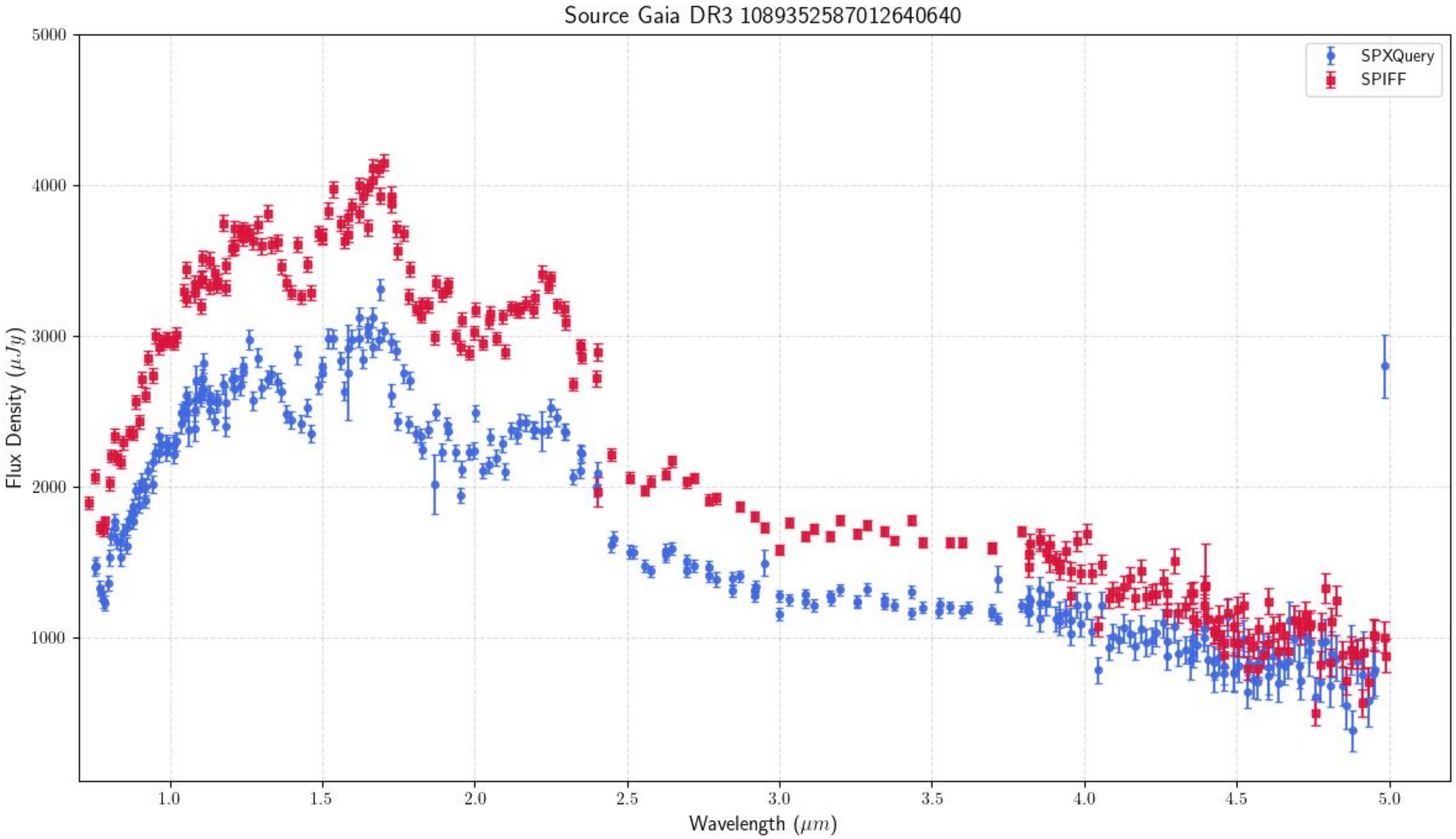
SPXQuery
(High Speed, Bulk Extraction)

- **https://github.com/WenkeRen/spxquery**
- Automates the full pipeline from querying to photometry using **parallel processing**
- Downloads only localized image cutouts, **reducing storage footprint by ~99%.**
- **The Limitation**: It relies on simple **aperture photometry** (e.g., a **3-pixel / 9.3" radius**). While fast, it systematically misses faint light in the PSF wings/tails, underreporting total flux.

SPXQuery
(High Speed, Bulk Extraction)

- **https://github.com/jgagneastro/SPIFF**
- Uses the exact instrumental **SPHEREx PSF**
- Crucially accounts for target **proper motion** across different SPHEREx **epochs**
- **The Limitation**: It is **accurate but slow** for large catalogs, taking > 2 hours to extract a single source.

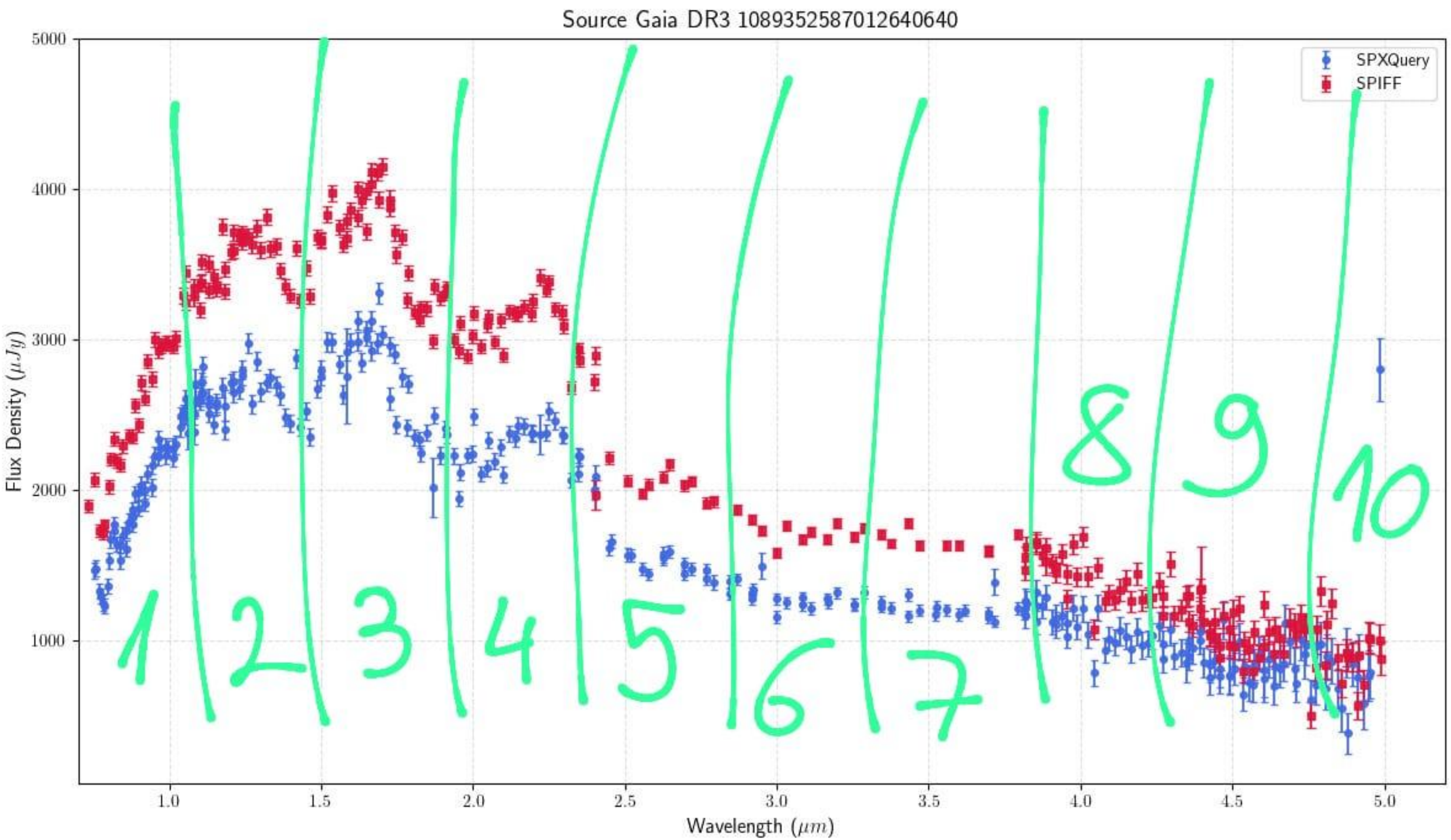
Spectral Extraction: SPXQuery vs. SPIFF



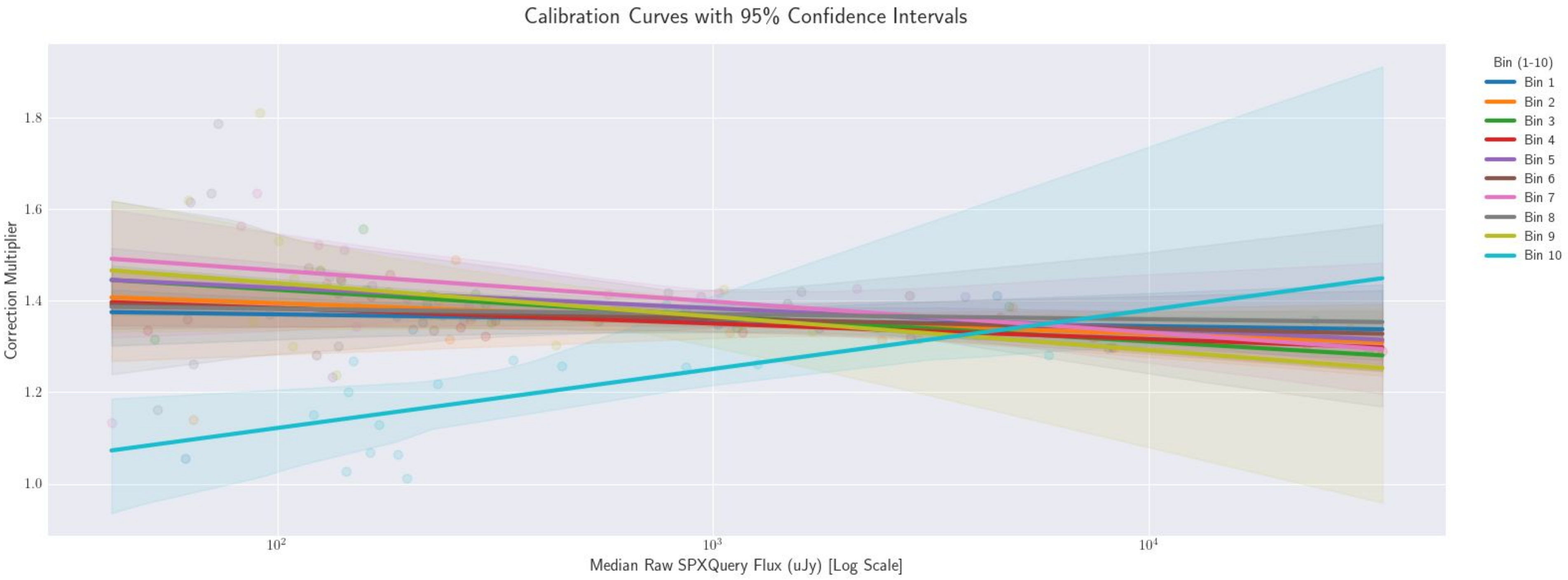
Empirical Calibration Logic

- **Problem**:
  - SPXQuery flux is consistently lower than SPIFF flux
- **Solution**:
  - We computed **correction factors** for multiple sources across various magnitudes and wavelengths
- **Implementation**:
  - The pipeline applies a magnitude-dependent correction across 10 wavelength bins
- **Result**:
  - SPXQuery aperture flux matches SPIFF PSF flux

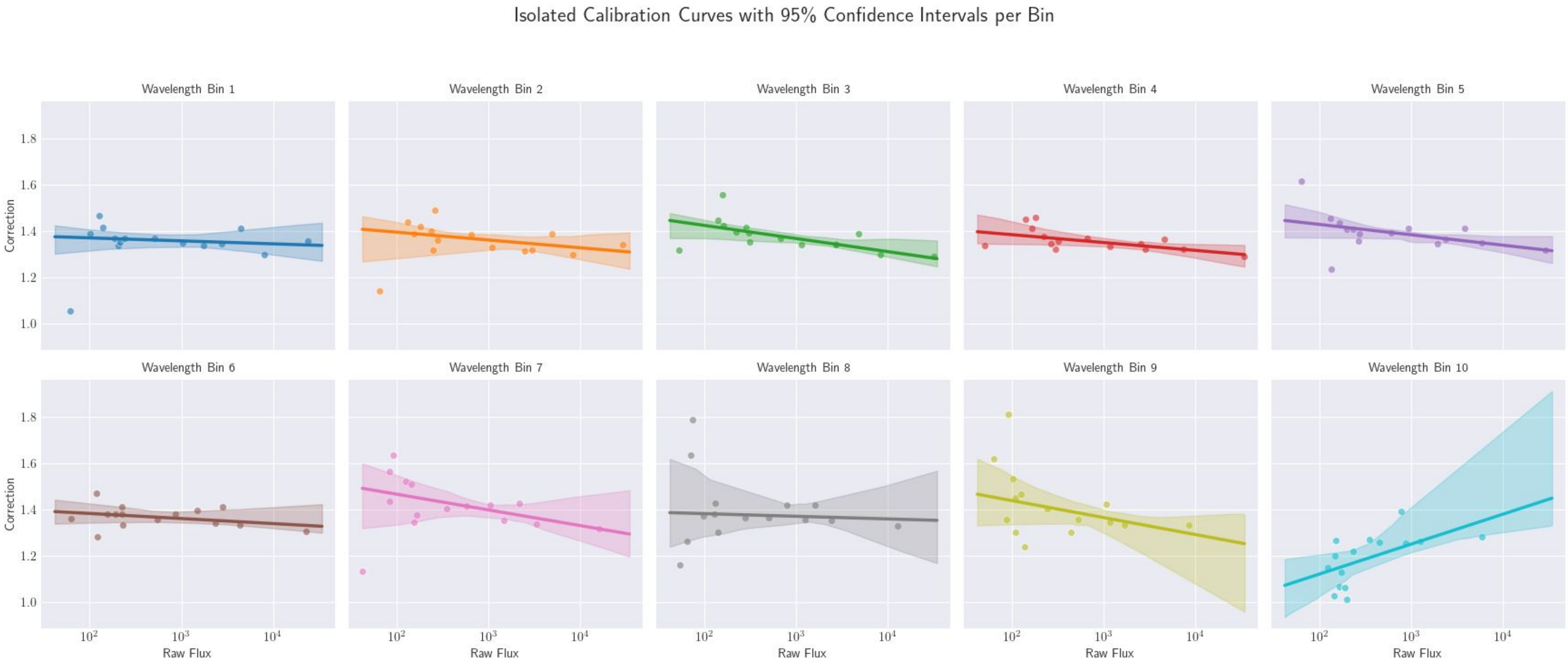
Empirical Calibration Logic



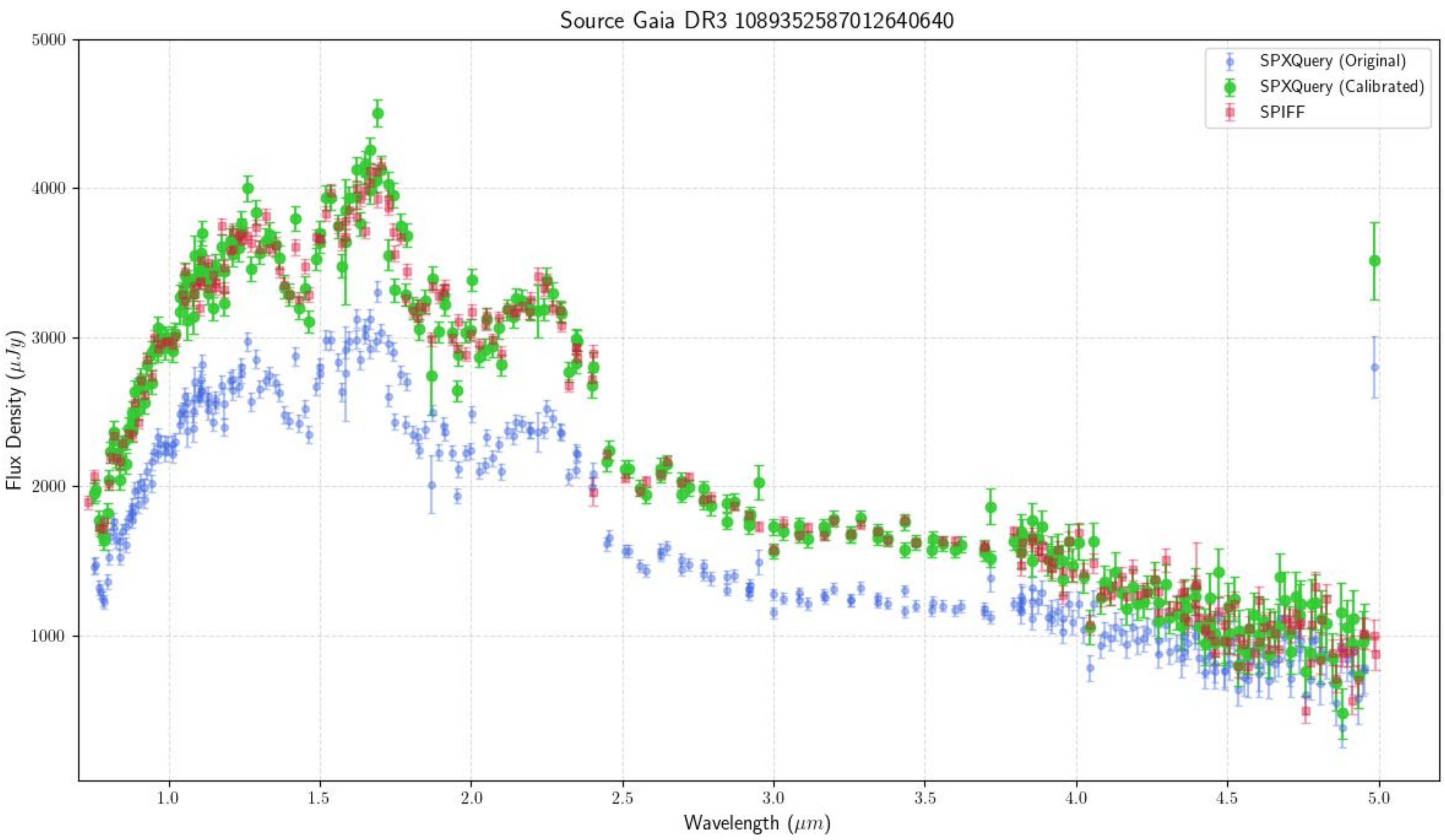
Empirical Calibration Logic



Empirical Calibration Logic



Calibration Visualized



Summary

- **Clean first, query second**
  - Use the 9.3" radius to match the extraction aperture
- **Code:**
  - https://github.com/andrijazupic/SPHEREx_pipeline